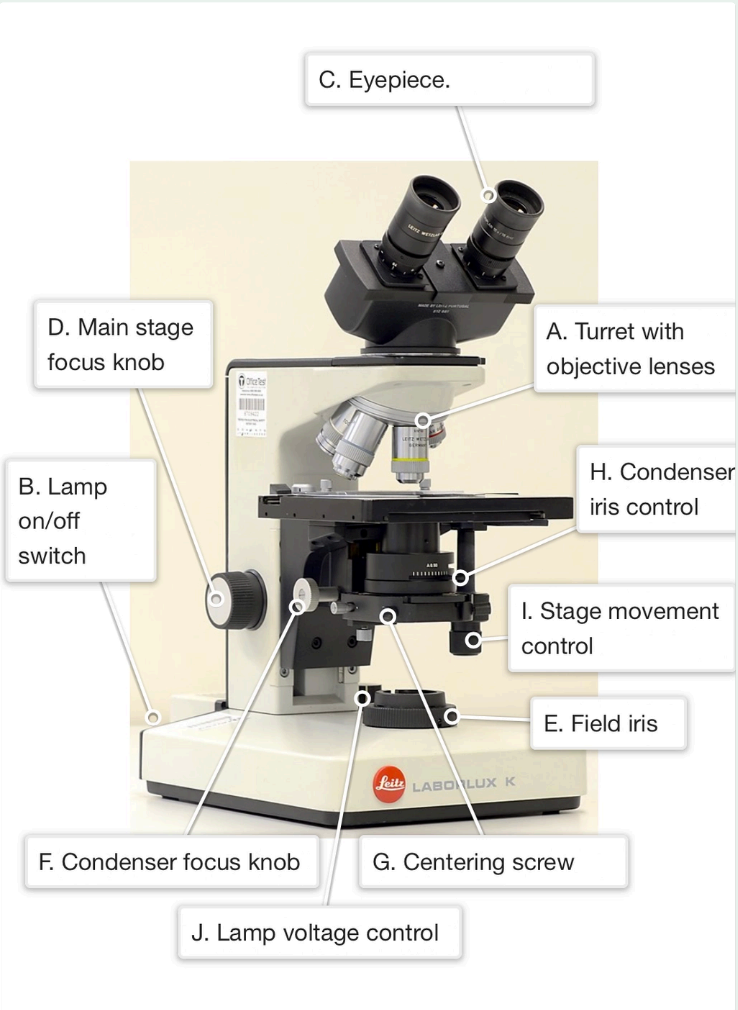
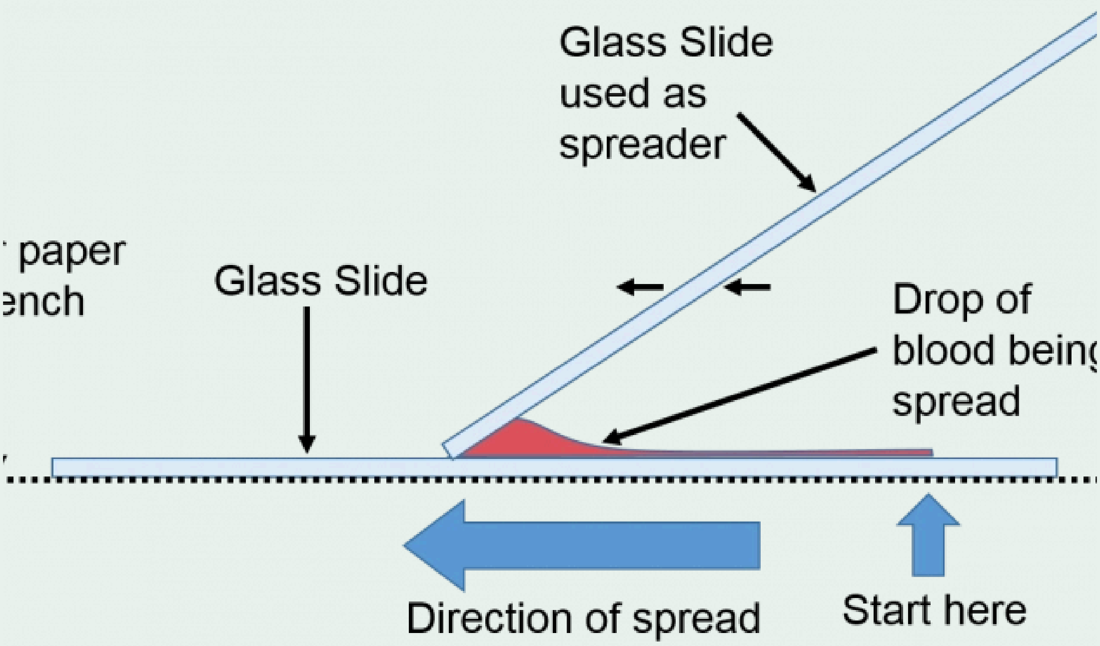
3.6 Diagnostic Role of the Blood Film

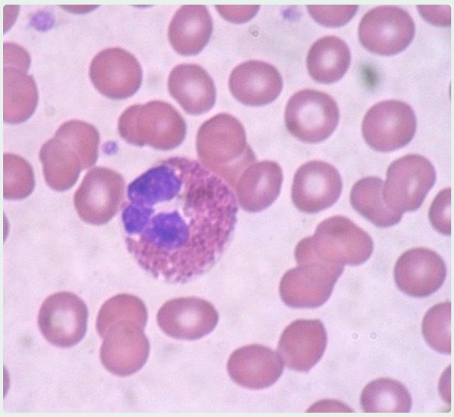
- Which stain, used in this practical, is the most frequently used stain for visualisation under microscopes?
  - **Haematoxylin & Eosin (H&E)**
  - This stain is used to view cells at visible wavelengths
- What is the role of haematoxylin in the stain?
  - It is blue, **binds DNA** and **shows up the nucleus.**
- What is the role of eosin in the stain?
  - It is pink, **binds protein components**, particularly in the **cytoplasm**
- Image showing a typical light microscope.



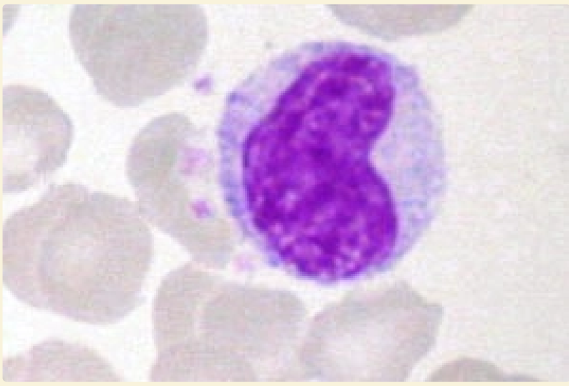
- Describe the steps involved in setting up a microscope correctly.
  - Preparation: Make sure the **microscope is set to the lowest magnification (4x)** and **ensure that the lamp is turned on**
  - **Step 1: Focus the image**
    - Use the main stage focus knob to make sure the image is in the right focus to see it clearly
  - **Step 2: Focus the condenser**
    - Close the field iris and adjust the condenser focus knob until you get a sharp image of the edge of the disc of light coming through the iris diaphragm
    - This should be roughly in the centre of the field of view – if it is not, adjust it with the centering screws
  - **Step 3: Adjust the field iris**
    - Open the field iris again and stop when the whole field of view is only just illuminated.
  - **Step 4: Adjust the condenser iris**
    - Start with the condenser iris fully open and then close it until it only just begins to darken the image.
  - **Examine:** Look at your specimen, **making any fine focus adjustments where necessary**
- Image showing how to spread blood across a microscope slide correctly.



- What is the function of the field iris of a microscope?
  - To **control the size of the "field" or area** that you are viewing under a microscope
- What is the function of the condenser of a microscope?
  - To allow a **uniform and maximum intensity of light** to be transferred **from the lamp to the area of the slide (field) that you are examining**
- What is special about the staining of white blood cells under Leishmann's stain?
  - **White blood cells** actually **stain blue**
- What properties are characteristic of eosinophils in identification under a microscope?
  - The bi-lobed nucleus and pink granules



- What property is characteristic of monocytes in identification under a microscope?
  - The **indented nucleus** is a good way of identifying a leukocyte as a monocyte.



▼ What properties are characteristic of basophils in identification under a microscope?

- The **blue stained granules** and **lobed nucleus** mark a leukocyte as a basophil.

▼ What properties are characteristic of lymphocytes in identification under a microscope?

- The **large round nucleus** and **relative lack of cytoplasm** marks a cell as a lymphocyte.

▼ What can happen to the cytology of neutrophils after a severe bacterial infection?

- The presence of **clear vacuoles within the cytoplasm** of the neutrophils can be seen

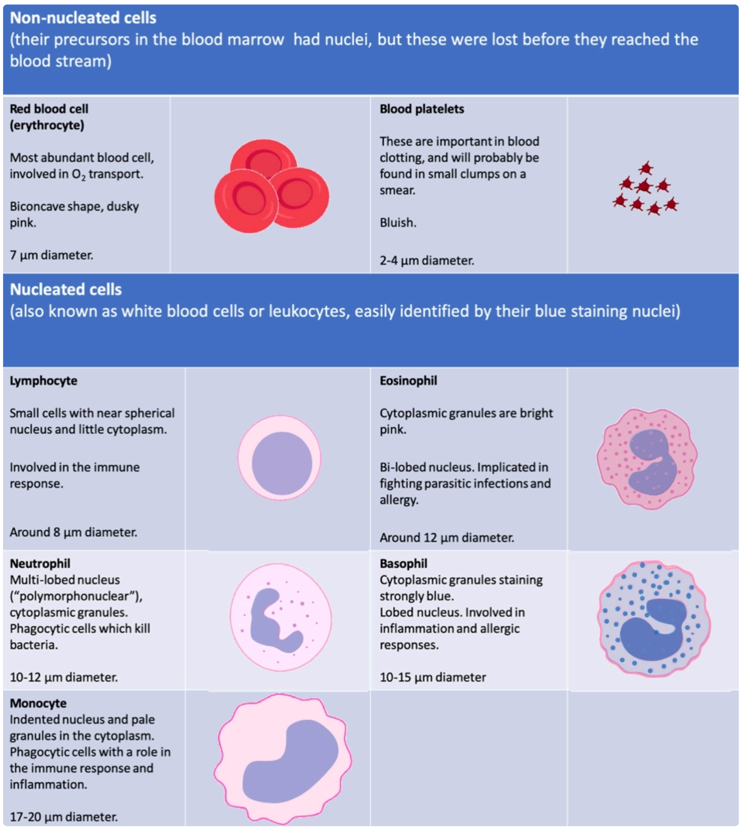
▼ What general chromosomal abnormality needs to occur for chronic myeloid leukaemia (CML) to be present?

- **Translocation** on the **Philadelphia chromosome**
    - i.e abnormal chromosome 22
- This causes the **BCR:ABL1 fusion gene** behind CML

▼ What is megaloblastic anaemia in terms of leukocytes?

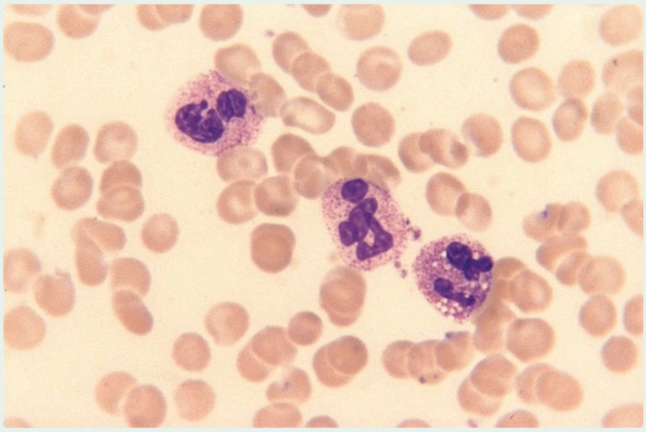
- Anaemia where the **nucleus is retarded (behind in development) in comparison to the cytoplasm**

▼ Image showing the different blood cells present in a blood film.

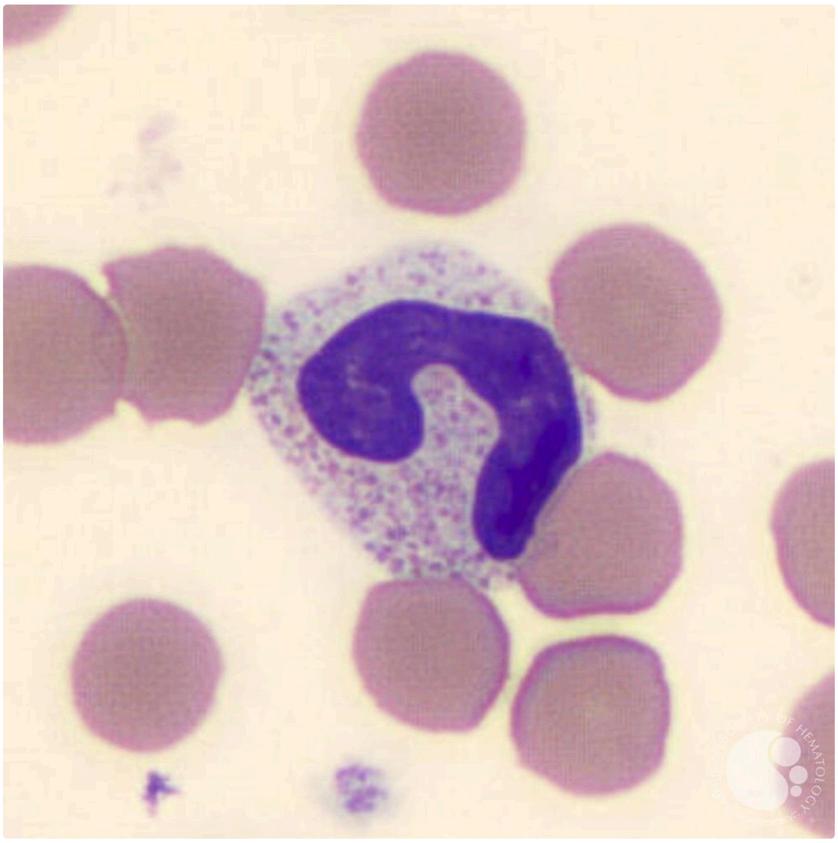


▼ What does being able to see clear vacuoles through a neutrophil suggest?

- **Bacterial infection**



- Image showing a band form neutrophil (nucleus has not yet lobulated).



- What genetic feature is essential in chronic myeloid leukaemia (CML)?
    - **BCR-ABL1 fusion gene**
        - Its presence can signify CML
- What sign in leukocytes can help us to identify megaloblastic anaemia?
    - **Hypersegmented nuclei** of neutrophils